# MSG No. 16.3 $P2'2'2$ [ Type III, orthorhombic ]

Table 1: Wyckoff site: `1a`, site symmetry: `2'2'2`

| No. | position | mapping |
|---|---|---|
| 1 | $[0, 0, 0]$ | `[1,2,3,4]` |

Table 2: Wyckoff site: `1b`, site symmetry: `2'2'2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, 0, 0\right]$ | `[1,2,3,4]` |

Table 3: Wyckoff site: `1c`, site symmetry: `2'2'2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, \frac{1}{2}, 0\right]$ | `[1,2,3,4]` |

Table 4: Wyckoff site: `1d`, site symmetry: `2'2'2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, 0, \frac{1}{2}\right]$ | `[1,2,3,4]` |

Table 5: Wyckoff site: `1e`, site symmetry: `2'2'2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, \frac{1}{2}, 0\right]$ | `[1,2,3,4]` |

Table 6: Wyckoff site: `1f`, site symmetry: `2'2'2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, 0, \frac{1}{2}\right]$ | `[1,2,3,4]` |

Table 7: Wyckoff site: `1g`, site symmetry: `2'2'2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, \frac{1}{2}, \frac{1}{2}\right]$ | `[1,2,3,4]` |

Table 8: Wyckoff site: `1h`, site symmetry: `2'2'2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | `[1,2,3,4]` |

Table 9: Wyckoff site: `2i`, site symmetry: `2'..`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ 0,\ 0]$ | `[1,3]` |
| 2 | $[-x,\ 0,\ 0]$ | `[2,4]` |

Table 10: Wyckoff site: `2j`, site symmetry: `2'..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ 0,\ \frac{1}{2}\right]$ | `[1,3]` |
| 2 | $\left[-x,\ 0,\ \frac{1}{2}\right]$ | `[2,4]` |

Table 11: Wyckoff site: `2k`, site symmetry: `2'..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ \frac{1}{2},\ 0\right]$ | `[1,3]` |
| 2 | $\left[-x,\ \frac{1}{2},\ 0\right]$ | `[2,4]` |

Table 12: Wyckoff site: `2l`, site symmetry: `2'..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[1,3]` |
| 2 | $\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[2,4]` |

Table 13: Wyckoff site: `2m`, site symmetry: `.2'.`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ y,\ 0]$ | `[1,4]` |
| 2 | $[0,\ -y,\ 0]$ | `[2,3]` |

Table 14: Wyckoff site: `2n`, site symmetry: `.2'.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ y,\ \frac{1}{2}\right]$ | `[1,4]` |
| 2 | $\left[0,\ -y,\ \frac{1}{2}\right]$ | `[2,3]` |

Table 15: Wyckoff site: `2o`, site symmetry: `.2'.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2},\ y,\ 0\right]$ | `[1,4]` |
| 2 | $\left[\frac{1}{2},\ -y,\ 0\right]$ | `[2,3]` |

Table 16: Wyckoff site: 2p, site symmetry: .2'.

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2},\ y,\ \frac{1}{2}\right]$ | [1,4] |
| 2 | $\left[\frac{1}{2},\ -y,\ \frac{1}{2}\right]$ | [2,3] |

Table 17: Wyckoff site: 2q, site symmetry: ..2

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ z]$ | [1,2] |
| 2 | $[0,\ 0,\ -z]$ | [3,4] |

Table 18: Wyckoff site: 2r, site symmetry: ..2

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2},\ 0,\ z\right]$ | [1,2] |
| 2 | $\left[\frac{1}{2},\ 0,\ -z\right]$ | [3,4] |

Table 19: Wyckoff site: 2s, site symmetry: ..2

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ \frac{1}{2},\ z\right]$ | [1,2] |
| 2 | $\left[0,\ \frac{1}{2},\ -z\right]$ | [3,4] |

Table 20: Wyckoff site: 2t, site symmetry: ..2

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$ | [1,2] |
| 2 | $\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$ | [3,4] |

Table 21: Wyckoff site: 4u, site symmetry: 1

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | [1] |
| 2 | $[-x,\ -y,\ z]$ | [2] |
| 3 | $[x,\ -y,\ -z]$ | [3] |
| 4 | $[-x,\ y,\ -z]$ | [4] |